CT EULER

- $e^{j\omega_0 t} = \cos(\omega_0 t) + j\sin(\omega_0 t)$
- $\cos(\omega_0 t) = \frac{1}{2}\left[e^{-j\omega_0 t} + e^{j\omega_0 t}\right]$
- $\sin(\omega_0 t) = \frac{j}{2}\left[e^{-j\omega_0 t} - e^{j\omega_0 t}\right]$

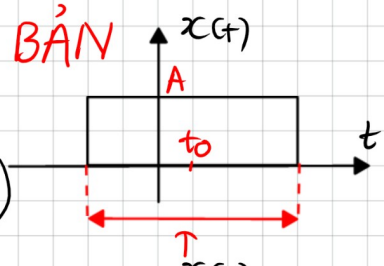
CT NĂNG LƯỢNG/CÔNG SUẤT

- $E_x = \int_{-\infty}^{+\infty} |x(t)|^2 \, dt$
- $P_y = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt$
- $P_y = \frac{1}{T} \int_{t_0}^{t_0+T} |x(t)|^2 dt$

TÍN HIỆU CƠ BẢN

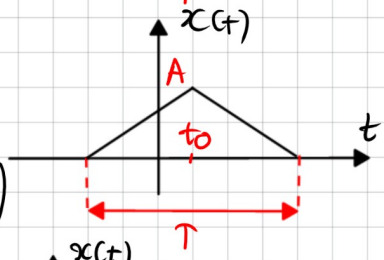
- Xung vuông

$x(t) = A \sqcap \left(\frac{t-t_0}{T}\right)$



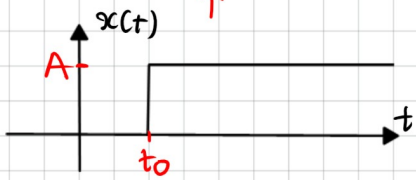
- Xung tam giác

$x(t) = A \wedge \left(\frac{t-t_0}{T/2}\right)$



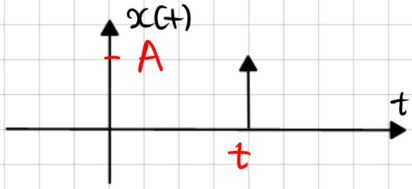
- Hàm bước

$x(t) = A.u(t-t_0)$



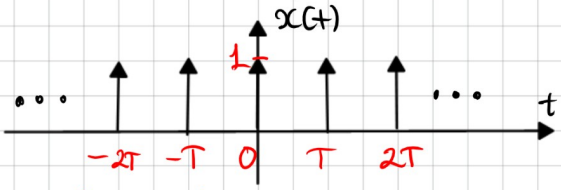
- Hàm dirac

$x(t) = A.\delta(t-t_0)$



- Phân bố lược

$x(t) = \frac{1}{T}\text{III}\left(\frac{t}{T}\right) = \sum_{n=-\infty}^{+\infty} \delta(t-nT)$



- Tính chất

$\int_{-\infty}^{+\infty} \delta(t)dt = 1$

$x(t) * \delta(t-t_0) = x(t-t_0)$

$x(t).\delta(t-t_0) = x(t_0).\delta(t-t_0)$

$x(t) * \frac{1}{T}\text{III}\left(\frac{t}{T}\right) = \sum_{n=-\infty}^{+\infty} x(t-nT)$

$x(t).\frac{1}{T}\text{III}\left(\frac{t}{T}\right) = \sum_{n=-\infty}^{+\infty} x(nT).\delta(t-nT)$

TÍNH CHẤT TÍN HIỆU

- Dịch chuyển:

$x(t-t_0)$: dời t_0 sang phải

$x(t+t_0)$: dời t_0 sang trái

- Lật (gấp): $x(-t)$
- Gấp + dịch chuyển : gấp trước, dời sau

$x(-t+t_0) = x(-(t-t_0))$

$x(-t-t_0) = x(-(t+t_0))$

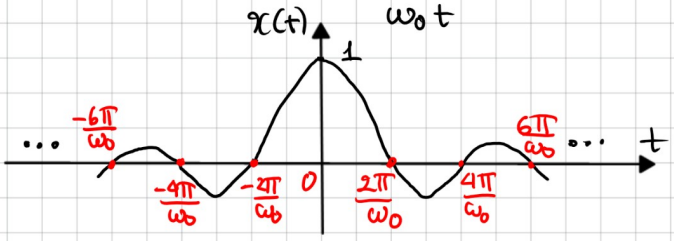
- Co / giãn : $x(\alpha.t) \to$ co

$(\alpha > 1)$ $x\left(\frac{t}{\alpha}\right) \to$ giãn

TÍN HIỆU CƠ BẢN (cont)

- Hàm Sa (sinc) $\omega_0 t = k\pi$

$x(t) = Sa(\omega_0 t) = \frac{\sin(\omega_0 t)}{\omega_0 t}$



- Hàm Sa^2 (sinc) $\omega_0 t = k\pi$

$x(t) = Sa^2(\omega_0 t) = \left[\frac{\sin(\omega_0 t)}{\omega_0 t}\right]^2$

TƯƠNG QUAN

• Tự tương quan:

$$\varphi_{xx}(\tau) = \int_{-\infty}^{\infty} x(t).\, x^*(t-\tau)dt \quad \text{(t/h NL)}$$

$$\psi_{xx}(\tau) = \frac{1}{T}\int_{t_0}^{t_0+T} x(t).\, x^*(t-\tau)dt \quad \text{(t/h CS)}$$

• Tính chất:

• $\varphi_{xx}(0) = E_x$ • $\psi_{xx}(0) = P_x$

• $\varphi_{xx}(\tau) = \varphi^*_{xx}(-\tau)$ • $\psi_{xx}(\tau) = \psi^*_{xx}(-\tau)$

• Tương quan chéo

$$\varphi_{xh}(\tau) = \int_{-\infty}^{\infty} x(t).\, h^*(t-\tau)dt \quad \text{(t/h NL)}$$

$$\varphi_{hx}(\tau) = \int_{-\infty}^{\infty} h(t).\, x^*(t-\tau)dt \quad \text{(t/h NL)}$$

$$\psi_{xh}(\tau) = \frac{1}{T}\int_{t_0}^{t_0+T} x(t).\, h^*(t-\tau)dt \quad \text{(t/h CS)}$$

$$\psi_{hx}(\tau) = \frac{1}{T}\int_{t_0}^{t_0+T} h(t).\, x^*(t-\tau)dt \quad \text{(t/h CS)}$$

• Tính chất:

• $\varphi_{xh}(\tau) = \varphi^*_{hx}(-\tau)$ • $\psi_{xh}(\tau) = \psi^*_{hx}(-\tau)$

TÍCH CHẬP

$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(t').\, \underbrace{h(t-t')}_{\text{lật + dịch}}dt'$$

• Tính chất:

$x(t) * h(t) = h(t) * x(t)$

$x(t)[h(t) + y(t)] = x(t) * h(t) + x(t) * y(t)$

BIẾN ĐỔI LAPLACE (cont)

• PP toán tử (phân số từng phần)

$$Y(s) = \frac{K(s)}{(s-a_1).(s^2+a_2^2).(s-a_3)^2 \ldots}$$

$$= \frac{A_0}{s-a_1} + \frac{A_1}{s-ja_2} + \frac{A_1^*}{s+ja_2} + \frac{A_2}{(s-a_3)^2} + \frac{A_3}{s-a_3}$$

• Với A_0, A_1, A_2: $\lim_{s \to (\text{giá trị mẫu} = 0)} [Y(s).(\text{mẫu})]$

• Với $A_3 = \lim_{s \to a_3} \frac{d}{ds}[Y(s).(s-a_3)]$

BIẾN ĐỔI LAPLACE

• Bảng công thức

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$t.u(t)$	$\frac{1}{s^2}$
$\frac{t^2}{2}u(t)$	$\frac{1}{s^3}$
$e^{-\alpha t}.u(t)$	$\frac{1}{s+\alpha}$
$t^n.e^{-\alpha t}.u(t)$	$\frac{n!}{(s+\alpha)^{n+1}}$
$\sin(\omega_0 t).u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$
$\cos(\omega_0 t).u(t)$	$\frac{s}{s^2+\omega_0^2}$
$e^{-\alpha t}.\cos(\omega_0 t).u(t)$	$\frac{s+\alpha}{(s+\alpha)^2+\omega_0^2}$
$e^{-\alpha t}.\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(s+\alpha)^2+\omega_0^2}$
$2r.\cos(\omega_0 t+\varphi).u(t)$	$\frac{r\angle\varphi}{s-j\omega_0} + \frac{r\angle-\varphi}{s+j\omega_0}$
$2r.e^{-\alpha t}.\cos(\omega_0 t+\varphi).u(t)$	$\frac{r\angle\varphi}{(s+a)-j\omega_0} + \frac{r\angle-\varphi}{(s+a)+j\omega_0}$

• Tính chất:

• Dịch chuyển miền tg:

$$f(t \pm t_0) \longleftrightarrow F(s).e^{\pm s.t_0}$$

• Dịch chuyển miền s:

$$f(t).e^{\mp s_0 t} \longleftrightarrow F(s \pm s_0)$$

• Tích chập: $x(t) * h(t) \longleftrightarrow X(s).H(s)$

• Đạo hàm:

$$\frac{d}{dt}f(t) \longleftrightarrow s.F(s) - F(0)$$

• Tích phân:

$$\int_{-\infty}^{t} f(\tau)d\tau \longleftrightarrow \frac{1}{s}F(s)$$

BIẾN ĐỔI FOURIER

• Bảng công thức:

$f(t)$	$F(\omega)$
1	$2\pi.\delta(\omega)$
$A\Pi\left(\frac{t}{T}\right)$	$AT\,Sa\left(\frac{\omega T}{2}\right)$
$A\Lambda\left(\frac{t}{T}\right)$	$AT\,Sa^2\left(\frac{\omega T}{2}\right)$
$Sa(\omega_0 t)$	$\frac{\pi}{\omega_0}\Pi\left(\frac{\omega}{2\omega_0}\right)$
$Sa^2(\omega_0 t)$	$\frac{\pi}{\omega_0}\Lambda\left(\frac{\omega}{2\omega_0}\right)$
$\cos(\omega_0 t)$	$\pi[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)]$
$\sin(\omega_0 t)$	$\pi j[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)]$
$\frac{1}{T}III\left(\frac{t}{T}\right)$	$III\left(\frac{\omega}{\omega_0}\right)=\omega_0\sum_{n=-\infty}^{+\infty}\delta(\omega-n\omega_0)$

• Tính chất:

• Dịch chuyển miền t:

$$f(t\pm t_0)\longleftrightarrow F(\omega).e^{\pm jt_0\omega}$$

• Dịch chuyển miền ω:

$$f(t).e^{\mp j\omega_0 t}\longleftrightarrow F(\omega\pm\omega_0)$$

• Tích chập:

$$x(t)*h(t)\longleftrightarrow X(\omega).H(\omega)$$

• Tích nhân

$$x(t).h(t)\longleftrightarrow\frac{1}{2\pi}X(\omega)*H(\omega)$$

• Định lý điều chế:

$$x(t).\cos(\omega_0 t)\longleftrightarrow\frac{1}{2}[X(\omega+\omega_0)+X(\omega-\omega_0)]$$

$$x(t).\sin(\omega_0 t)\longleftrightarrow\frac{j}{2}[X(\omega+\omega_0)+X(\omega-\omega_0)]$$

• Tương quan:

$$\varphi_{xy}(\tau)\longleftrightarrow X(\omega).Y^*(\omega)$$

• Định lý Parseval

+) Mật độ phổ NL:

$$\Phi(\omega)=|X(\omega)|^2$$

+) Năng lượng

$$E_x=\int_{-\infty}^{+\infty}\Phi(\omega)d\omega=\frac{1}{2\pi}\int_{-\infty}^{+\infty}|X(\omega)|^2d\omega$$

+) Mật độ phổ CS:

$$\Psi(\omega)=2\pi\sum_{n=-\infty}^{+\infty}|X_n|^2\delta(\omega-n\omega_0)$$

+) Công suất:

$$P_y=\sum_{n=-\infty}^{+\infty}|X_n|^2$$